

FedAKD: Federated Adaptive Knowledge Distillation via Global Knowledge Calibration and Decoupling

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Abstract

The rapid growth of Web, mobile, and Web of Things (WoT) applications increasingly relies on Federated Learning (FL) to enable privacy-preserving intelligence across distributed and heterogeneous devices. However, the global model in FL often suffers from performance degradation due to global knowledge forgetting caused by highly non-IID client data. While client-side Knowledge Distillation (KD) has emerged as a promising paradigm for transferring global knowledge to local models, existing approaches inadvertently distort global knowledge through local data priors, fundamentally limiting their effectiveness in real-world Web and mobile environments. We further find that this knowledge forgetting is asymmetric, with locally under-represented classes suffering significantly more severe performance degradation than over-represented ones. While the conventional KD inherently couples these two distinct kinds of knowledge, thereby hindering the targeted preservation of critical global knowledge for under-represented classes. To address these challenges, we propose Federated Adaptive Knowledge Distillation (FedAKD), a novel framework that adaptively calibrates and decouples global knowledge according to the local data distribution, enabling precise and flexible distillation across heterogeneous clients. Extensive experiments on benchmark datasets and diverse data heterogeneity settings demonstrate that FedAKD achieves state-of-the-art performance, offering a scalable and sustainable solution for Web, mobile, and WoT intelligence deployment.

CCS Concepts

• **Computing methodologies** → **Cooperation and coordination**; *Mobile agents*.

Keywords

Federated Learning, Knowledge Distillation, Non-IID, Knowledge Calibration, Knowledge Decoupling

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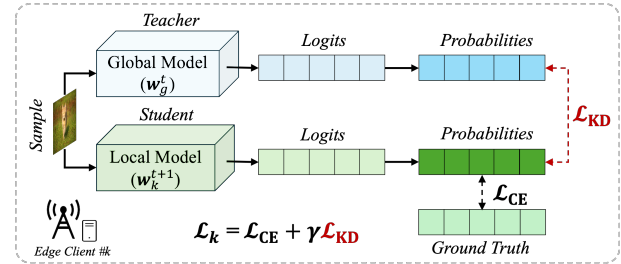


Figure 1: The framework of client-side KD in FL.

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1 Introduction

Federated Learning (FL)[16] is a distributed training paradigm that enables multiple clients to train a global model without aggregating their data, thereby preserving user privacy [16, 23, 28]. With the rapid proliferation of Web, mobile, and Web of Things (WoT) applications, FL has become an attractive framework for delivering intelligent services across highly distributed and resource-constrained devices. Despite its potential, FL encounters significant challenges in practice, primarily due to the non-Independent and Identically Distributed (non-IID) nature of the data, referred to as data heterogeneity, such as label distribution skew, feature distribution skew, and quantity skew [13, 30]. In this paper, we focus on a prevalent form of non-IID data, i.e., label distribution skew. This phenomenon manifests as imbalanced local datasets containing both over-represented (majority) classes with more samples and under-represented (minority) classes with fewer samples [3, 12, 31]. Such skewed label distribution across clients not only hinders the global model's convergence but also degrades its generalization capability [3, 12, 13, 30, 31].

To address this fundamental challenge, recent studies [1, 4, 5, 9, 12, 14, 18, 20, 25, 27, 29] have identified catastrophic forgetting [19] of global knowledge as a critical bottleneck in non-IID FL, where the global model gradually adapting to the specific data distribution of certain clients while forgetting knowledge from the data of other clients. To counteract this issue, client-side Knowledge Distillation (KD) [3–5, 9, 12, 29] has emerged as a promising paradigm. As shown in Figure 1,¹ the output probability (soft label)

¹Without loss of generality, ignore the distillation temperature.

generated by the latest global model $\mathbf{w}_g^t$ serve as the global knowledge to guide the training of the local model $\mathbf{w}_k^{t+1}$. This process establishes a distillation regularization mechanism for the local model, enabling local models to preserve global knowledge learned from the federated consensus while adapting to client-specific data distributions.

Although the client-side KD approach is practical, we reveal two fundamental limitations in existing approaches:

- Existing client-side KD methods implicitly assume that local student models can faithfully replicate the unbiased knowledge representations from the global teacher model. However, we demonstrate that the inherent discrepancy between local and global prior distributions compels the student model to compensate for this prior discrepancy by distorting its likelihood estimation in the feature space, resulting in the acquisition of distorted global knowledge contaminated by local biases (Section 3.1).
- We uncover an asymmetric knowledge forgetting phenomenon: local under-represented classes experience an average of $1.37\times$ more severe knowledge forgetting than local over-represented classes in typical non-IID scenarios (Figure 2). However, conventional KD methods inadvertently entangle the global knowledge of under- and over-represented classes through undifferentiated KL divergence objectives, thereby preventing the targeted preservation of the under-represented class's global knowledge (Section 3.2).

To address these challenges, we propose Federated Adaptive Knowledge Distillation (FedAKD). First, we perform distribution-aware calibration on the teacher model's logits to mitigate discrepancies between local and global prior distributions, followed by masking the teacher's and student's ground-truth class logits. Subsequently, we adaptively decompose the calibrated and masked output probabilities into Over-represented Class Knowledge (OCK), Under-represented Class Knowledge (UCK), and Group Relational Knowledge (GRK), then conduct KD on each of them, thereby achieving discriminative preservation of global knowledge. Experimental results confirm that FedAKD achieves state-of-the-art performance across diverse non-IID scenarios. The main contributions of this study can be summarized as follows.

- **Theoretical characterization of global knowledge distortion.** We establish a Bayesian theoretical framework revealing how local biases induce biased global knowledge through implicit prior encoding in federated KD, which explains why conventional KD fails to preserve genuine global feature-class knowledge under heterogeneous priors.
- **Quantification of asymmetric knowledge forgetting.** We propose a quantitative framework for class-wise forgetting analysis in FL, identifying that under-represented classes suffer an average of $1.37\times$ more severe forgetting than over-represented classes, revealing the critical need for discriminative knowledge preservation strategies.
- **Calibrated and decoupled federated KD framework.** FedAKD featuring dual innovations: (i) Distribution-aware teacher logit calibration that removes bias while preserving transferable global knowledge; (ii) Decoupled distillation

with independent control of GRK, OCK, and UCK distillation, enabling targeted mitigation of forgetting.

2 Related Work

2.1 Knowledge Forgetting in FL

The catastrophic forgetting [19] that occurs in FL is caused by local model updates on heterogeneous data, leading the global model to gradually adapt to the specific data distribution of certain clients while forgetting knowledge from the data of other clients. To address this issue, some replay-based [18, 25] and regularization-based [1, 14, 20, 27] approaches have been proposed. For example, Pennisi et al. [18] used memory buffers of synthetic privacy-preserving samples and interleaving training on local data and on buffer data. Shoham et al. [20] introduced an elastic weight consolidation penalty term to the loss function to avoid forgetting previously learned knowledge. Unlike these studies, we primarily employ KD to mitigate the problem of forgetting.

2.2 Client-Side KD in FL

Recent studies [3–5, 9, 12, 29] have introduced KD into local training to address forgetting. FCCL [5] and FedGKD [29] adopted KD in local updating, but they lack specialized class distillation strategies tailored for FL processes, where non-IID data may compromise the effectiveness of conventional knowledge transfer mechanisms. In contrast, FedSSD [4] selectively distilled teacher knowledge based on output confidence, FedNTD [9] restricted KD to non-true class outputs, FedLMD [12] prioritized minority class preservation, while FedVLS [3] leveraged KD to retain vacant class knowledge. However, these approaches fail to address the inherent bias in local data distributions that induces student models to assimilate biased global knowledge. Moreover, these studies overlook the fundamental distinction between over- and under-represented classes in terms of their non-uniform knowledge forgetting rates during FL. Although FedLMD [12] advocated exclusive distillation of minority classes' knowledge, this rigid strategy neglects the problem of knowledge forgetting in over-represented classes. Therefore, we aim to resolve the above dual challenges simultaneously.

3 Revisting KD in FL

3.1 Knowledge Bias of the Global Teacher Model

Within the probabilistic interpretation framework of classification models, the output probability can be mathematically formulated as a parametric estimation of the posterior probability $P(y|\mathbf{x}; \theta)$. By minimizing $\mathcal{L}_{\text{CE}} = -\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [\log P(y|\mathbf{x}; \theta)]$ (cross-entropy loss function), the model essentially performs maximum likelihood estimation, with the objective being equivalent to minimizing the classification error rate $\mathbb{P}_{(\mathbf{x}, y) \sim \mathcal{D}} (y \neq \arg \max_{y'} P(y'|\mathbf{x}; \theta))$.

According to Bayes' theorem, the posterior probability can be decomposed as $P(y|\mathbf{x}) = \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})}$, where $P(\mathbf{x}) = \sum_{y'} P(\mathbf{x}|y')P(y')$. Since the softmax cross-entropy is usually chosen as the surrogate loss function, the probability $P(y|\mathbf{x}) \propto e^{z_y}$ is regarded as the estimate of $P(y|\mathbf{x})$ [2, 17, 31], where z_y is the logit preceding the softmax layer in neural networks. Therefore, $P(y|\mathbf{x})$ can be expressed

as follows.

$$P(y|\mathbf{x}) = \frac{e^{\ln P(\mathbf{x}|y) + \ln P(y)}}{\sum_{y'} e^{\ln P(\mathbf{x}|y') + \ln P(y')}} \propto \frac{e^{z_y}}{\sum_{y'} e^{z_{y'}}}. \quad (1)$$

This indicates the optimal classifier depends exclusively on the sum of the log-likelihood $\ln P(\mathbf{x}|y)$ and log-prior $\ln P(y)$, and provides a theoretical foundation for the architectural design of logits z_y that can be represented as $z_y \propto \ln P(\mathbf{x}|y) + \ln P(y)$. Thus, during the local KD phase in FL, the global teacher model $\mathcal{T}$ approximately learns a globally distribution $P_g^{\mathcal{T}}(y)$ through periodic federated aggregation of client models,² and its logit decomposition can be expressed as $z_y^{\mathcal{T}} \propto \ln P_g(\mathbf{x}|y; \theta^{\mathcal{T}}) + \ln P_g^{\mathcal{T}}(y)$. In contrast, the local student model $\mathcal{S}$ is trained under local distribution $P_k^{\mathcal{S}}(y) \neq P_g^{\mathcal{T}}(y)$, its logits implicitly incorporate the local data bias, which can be expressed as $z_y^{\mathcal{S}} \propto \ln P_k(\mathbf{x}|y; \theta^{\mathcal{S}}) + \ln P_k^{\mathcal{S}}(y)$. The objective of client-side KD is to enforce alignment between the local posterior distribution and the global posterior distribution, i.e., $P_k^{\mathcal{S}}(y|\mathbf{x}) \approx P_g^{\mathcal{T}}(y|\mathbf{x})$, which implicitly requires logit matching $z_y^{\mathcal{S}} \approx z_y^{\mathcal{T}}$:

$$\begin{aligned} \ln P_k(\mathbf{x}|y; \theta^{\mathcal{S}}) + \ln P_k^{\mathcal{S}}(y) &\approx \ln P_g(\mathbf{x}|y; \theta^{\mathcal{T}}) + \ln P_g^{\mathcal{T}}(y) \\ \Rightarrow \ln P_k(\mathbf{x}|y; \theta^{\mathcal{S}}) &\approx \ln P_g(\mathbf{x}|y; \theta^{\mathcal{T}}) + \ln P_g^{\mathcal{T}}(y) - \ln P_k^{\mathcal{S}}(y). \end{aligned} \quad (2)$$

Given that the local prior $P_k^{\mathcal{S}}(y)$ is intrinsically determined by the client's data distribution, the student model must compensate for prior discrepancy ($\ln P_g^{\mathcal{T}}(y) - \ln P_k^{\mathcal{S}}(y)$) by adjusting the likelihood term $\ln P_k(\mathbf{x}|y; \theta^{\mathcal{S}})$. While this mechanism functionally mimics the global posterior, it distorts the true feature-class relationships ($P_k(\mathbf{x}|y) \neq P_g(\mathbf{x}|y)$), embedding local biases into the learned representations. Thus, the acquired knowledge is distorted global knowledge that preserves task performance locally but sacrifices cross-client generalizability due to feature-space shifts. This underscores the need for debiasing techniques like prior calibration of the teacher logit (Section 4.2).

3.2 Knowledge Forgetting and Coupling of the Global Teacher model

3.2.1 Global knowledge forgetting in federated learning. Previous studies [9, 21] utilized the Backward Transfer (BwT) idea to quantify the cumulative forgetting effect on each class c across all FL rounds, as follows:

$$\mathcal{F} = \frac{1}{C} \sum_{c=1}^C \max_{t \in \{1, 2, \dots, T-1\}} (\mathcal{A}_c^t - \mathcal{A}_c^T). \quad (3)$$

where $\mathcal{A}_c^t$ denotes the accuracy of the global model for class c at round t , and T denotes the total number of communication rounds.

$\mathcal{F}$ captures the average difference between the peak accuracy and the final accuracy for each class after FL, thereby reflecting the performance fluctuations of the global model over all historical rounds. However, it neglects the heterogeneous forgetting degrees across different classes in FL. For example, under-represented classes with lower local data proportions may be more prone to knowledge forgetting due to insufficient representation during local

model training. Therefore, we formally define **over-** and **under-represented** classes based on the local distributions, refining the granularity and interpretability of forgetting analysis in FL.

Definition 3.1. Consider a C -class classification task, and each client k 's local dataset $\mathcal{D}_k$ contain n_k training samples. For client k , the class distribution vector $\mathbf{d}_k = [d_k^{(1)}, d_k^{(2)}, \dots, d_k^{(C)}]$ is defined using a binary indicator:

$$d_k^{(i)} = \begin{cases} 0, & \text{if } \Delta^{(i)} \geq \nu, \\ 1, & \text{if } \Delta^{(i)} < \nu. \end{cases} \quad (4)$$

$$\Delta^{(i)} = \frac{p_k^{(i)} - p}{p}, p_k^{(i)} = \frac{n_k^{(i)}}{n_k}, \quad \text{and} \quad p = \frac{1}{C}. \quad (5)$$

$\nu \geq 0$ is a threshold parameter, with $d_k^{(i)} = 0$ marking over-represented classes ($y_i \in \mathbf{y}_k^{\text{over}}$) and $d_k^{(i)} = 1$ indicating under-represented classes ($y_i \in \mathbf{y}_k^{\text{under}}$).

Regarding the range of values for ν , we have the theorem 3.2:³

THEOREM 3.2. Given a significance level δ (e.g., $\delta = 0.05$), if the local proportion is significantly higher than the average proportion, the theoretical lower bound for ν is: $\nu \geq z_{\delta} \cdot \sqrt{\frac{1-p}{n_k p}}$, where z_{δ} is the upper δ -quantile of the standard normal distribution (e.g., $z_{0.05} \approx 1.645$).

Considering the over- and under-represented classes, we can define the maximum performance drop for a class c as follows:

$$\Delta \mathcal{A}_k^{(c)} = \max_{t \in \{1, 2, \dots, T-1\}} (\mathcal{A}_k^t(\mathbf{w}_g^t) - \mathcal{A}_k^T(\mathbf{w}_g^T)). \quad (6)$$

Using this term, we can express the knowledge forgetting of all clients for over- and under-represented classes more concisely as follows:

$$\mathcal{F}_{\text{under}} = \frac{1}{m} \sum_{k=1}^m \frac{\sum_{c=1}^C d_k^{(c)} \cdot \Delta \mathcal{A}_k^{(c)}}{\|\mathbf{d}_k\|_1}, \quad (7)$$

$$\mathcal{F}_{\text{over}} = \frac{1}{m} \sum_{k=1}^m \frac{\sum_{c=1}^C (1 - d_k^{(c)}) \cdot \Delta \mathcal{A}_k^{(c)}}{C - \|\mathbf{d}_k\|_1}. \quad (8)$$

where m represents the number of clients participating in FL, and $\mathbf{w}_t^g$ is the initial global model sent to clients at round t .

The phenomenon of knowledge forgetting in FedAvg [16], quantified via $\mathcal{F}$, $\mathcal{F}_{\text{over}}$, and $\mathcal{F}_{\text{under}}$ is shown in Figure 2.⁴ The results indicate that as the heterogeneity of client data increases, the degree of knowledge forgetting $\mathcal{F}$ becomes more severe. Notably, under-represented classes ($\mathcal{F}_{\text{under}}$) experience an average of 1.37× more severe knowledge forgetting than over-represented classes ($\mathcal{F}_{\text{over}}$) under equivalent heterogeneity conditions. This suggests that we should take effective measures to deal with the two types of knowledge forgetting problems separately.

³The proof is given in Appendix A. This study adopts the theoretical minimum value of ν ($\delta = 0.05$) by default.

⁴The experimental setup was consistent with that of the previous study [9].

²Without loss of generality, assume all clients participate.

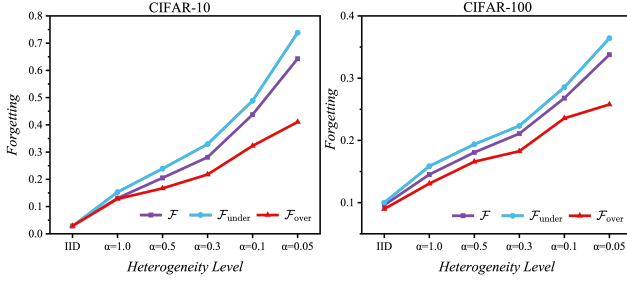


Figure 2: Knowledge forgetting in the global server model under varying levels of data heterogeneity (α) for CIFAR-10 and CIFAR-100 datasets. The parameter α of the Dirichlet prior $\text{Dir}(\alpha)$ governs the heterogeneity of local data distributions, where smaller α values induce stronger data heterogeneity.

3.2.2 Knowledge decoupling of the global model. As empirically validated in Figure 2, local under- and over-represented classes exhibit divergent magnitudes of knowledge forgetting during FL, which indicates that under-represented classes require a stronger intensity of KD. However, traditional KD methods based on Kullback-Leibler (KL) divergence tend to suppress the learning of under-represented classes' knowledge of the global model. To illustrate this issue, we decouple the output of the teacher model into over-represented class knowledge (OCK), under-represented class knowledge (UCK), and group relational knowledge (GRK) as follows.

Definition 3.3. Assume that the first l classes of client k are over-represented, and the remaining $o = C - l$ classes are under-represented.⁵ Define $\hat{\mathbf{p}}_{\text{over}} = [\hat{p}_1, \hat{p}_2, \dots, \hat{p}_l] \in \mathbb{R}^{1 \times l}$ as the over-represented class knowledge, $\check{\mathbf{p}}_{\text{under}} = [\check{p}_1, \check{p}_2, \dots, \check{p}_o] \in \mathbb{R}^{1 \times o}$ as the normalized under-represented class knowledge, and $\mathbf{p}_{\text{group}} = [p_{\text{over}}, p_{\text{under}}] \in \mathbb{R}^{1 \times 2}$ as group relational knowledge. The probabilities are computed as:

$$\begin{aligned} \hat{p}_i &= \frac{\exp(z_i)}{\sum_{j=1}^l \exp(z_j)}, & \check{p}_i &= \frac{\exp(z_{i+l})}{\sum_{j=l+1}^C \exp(z_j)}, \\ p_{\text{over}} &= \frac{\sum_{i=1}^l \exp(z_i)}{\sum_{j=1}^C \exp(z_j)}, & p_{\text{under}} &= \frac{\sum_{i=l+1}^C \exp(z_i)}{\sum_{j=1}^C \exp(z_j)}. \end{aligned} \quad (9)$$

where z_i represents the logit output of the i -th class by the model.

OCK ($\hat{\mathbf{p}}_{\text{over}}$) and UCK ($\check{\mathbf{p}}_{\text{under}}$) represent the model's normalized confidence distributions for the over- and under-represented class groups, respectively. GRK ($\mathbf{p}_{\text{group}}$) characterizes the model's holistic inter-group relational confidence between over- and under-represented classes, encapsulating their relative significance from a global perspective.

Based on Definition 3.3, we decouple and reconstruct the KD loss based on KL divergence as the following proposition.⁶

PROPOSITION 3.4. Let the teacher model and student model be denoted by $\mathcal{T}$ and $\mathcal{S}$, respectively. The distillation loss based on KL divergence can be decoupled into the following three components:

⁵The Softmax function and KL divergence exhibit permutation invariance with respect to class ordering.

⁶The proof is given in Appendix B.

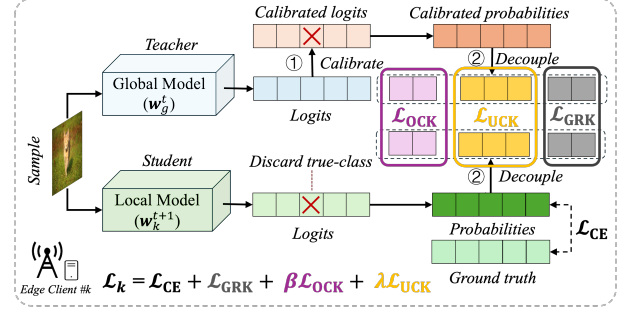


Figure 3: The client-side training framework of FedAKD.

$$\begin{aligned} \mathcal{L}_{KD} &= KL(\mathbf{p}^{\mathcal{T}} || \mathbf{p}^{\mathcal{S}}) = \underbrace{KL(\mathbf{p}_{\text{group}}^{\mathcal{T}} || \mathbf{p}_{\text{group}}^{\mathcal{S}})}_{\text{GRK distillation}} \\ &\quad + \underbrace{p_{\text{over}}^{\mathcal{T}} \cdot KL(\hat{\mathbf{p}}_{\text{over}}^{\mathcal{T}} || \hat{\mathbf{p}}_{\text{over}}^{\mathcal{S}})}_{\text{OCK distillation}} + \underbrace{p_{\text{under}}^{\mathcal{T}} \cdot KL(\check{\mathbf{p}}_{\text{under}}^{\mathcal{T}} || \check{\mathbf{p}}_{\text{under}}^{\mathcal{S}})}_{\text{UCK distillation}}. \end{aligned} \quad (10)$$

As shown in Proposition 3.4, the OCK distillation and UCK distillation are intrinsically coupled through the relationship $p_{\text{over}}^{\mathcal{T}} = 1 - p_{\text{under}}^{\mathcal{T}}$. This coupling implies that the distillation intensity allocated to over- and under-represented classes is inherently governed by the teacher model's confidence in their respective group probabilities. While this formulation preserves the probabilistic consistency of knowledge transfer, the fixed proportionality $p_{\text{over}}^{\mathcal{T}} + p_{\text{under}}^{\mathcal{T}} = 1$ restricts adaptive balancing between OCK and UCK distillation. Ideally, the distillation weights should dynamically reflect the local data distribution (e.g., prioritizing UCK to mitigate forgetting) rather than being solely dictated by the teacher's confidence.

4 Federated Adaptive Knowledge Distillation

Motivated by the challenges discussed in Section 3, we propose Federated Adaptive Knowledge Distillation (FedAKD).

4.1 Framework

Figure 3 illustrates the client-side training framework of FedAKD. While preserving the global aggregation protocol of FedAvg [16], FedAKD introduces a calibrated and decoupled KD mechanism during local training. The workflow comprises two steps: First, the logits of the global teacher model are calibrated using the estimated local and global prior distributions, followed by masking the ground-truth class logits in both the teacher's and student's output. Subsequently, the calibrated and masked output probabilities are decoupled into OCK, UCK, and GRK, enabling the construction of distinct distillation loss terms for each component. The detailed procedure of our proposed FedAKD is summarized in Algorithm 1.

4.2 Adaptive Knowledge Calibration

As derived in Section 3.1, the global teacher model inherently encodes the global class prior $P_g^{\mathcal{T}}(y)$ within its logit space $z_y^{\mathcal{T}}$. Directly aligning student logits $z_y^{\mathcal{S}}$ with $z_y^{\mathcal{T}}$ under heterogeneous local priors $P_k^{\mathcal{S}}(y)$ forces the student model to distort its likelihood estimation

Algorithm 1 FedAKD - Federated Adaptive Knowledge Distillation

Require: Total rounds T , Number of clients K , Client sampling ratio R , Local epochs E , Learning rate η , Weights λ, β

- 1: Initialize global model $\mathbf{w}_g^0$ and global prior $\mathbf{P}_g$
- 2: **for** each round $t = 0$ to $T - 1$ **do**
- 3: Server selects clients $\mathcal{S}_t \subseteq \{1, \dots, K\}, |\mathcal{S}_t| = R \cdot K$
- 4: **for** each client $k \in \mathcal{S}_t$ **in parallel do**
- 5: Receive global model $\mathbf{w}_g^t$ and global prior $\mathbf{P}_g$ from server
- 6: Initialize model $\mathbf{w}_k^t \leftarrow \mathbf{w}_g^t$ and compute local prior $\mathbf{P}_k$
- 7: **for** each local epoch $e = 0$ to $E - 1$ **do**
- 8: **for** each mini-batch $(\mathbf{x}, y)$ from $\mathcal{D}_k$ **do**
- 9: // Forward pass
- 10: Compute teacher logits: $\mathbf{z}^T = f_{\mathbf{w}_g^t}(\mathbf{x})$
- 11: Calibrate logits: $\mathbf{z}^{\text{cal}} = \mathbf{z}^T + \ln \mathbf{P}_k - \ln \mathbf{P}_g$
- 12: Compute student logits: $\mathbf{z}^S = f_{\mathbf{w}_k^t}(\mathbf{x})$
- 13: Apply ground-truth masking on $\mathbf{z}^{\text{cal}}$ and $\mathbf{z}^S$
- 14: Compute soft distributions: $\mathbf{p}^{\text{cal}}, \mathbf{p}^S$
- 15: // Loss computation and model update
- 16: Compute total loss $\mathcal{L}_{\text{FedAKD}}$ as Eq. 13
- 17: Update local model: $\mathbf{w}_k^t \leftarrow \mathbf{w}_k^t - \eta \cdot \nabla \mathcal{L}_{\text{FedAKD}}$
- 18: **end for**
- 19: **end for**
- 20: Send updated model $\mathbf{w}_k^{t+1} \leftarrow \mathbf{w}_k^t$ to server
- 21: **end for**
- 22: Server aggregate: $\mathbf{w}_g^{t+1} \leftarrow \text{FedAvg}(\{\mathbf{w}_k^{t+1}\}_{k \in \mathcal{S}_t})$
- 23: **end for**
- 24: **return** Final global model $\mathbf{w}_g^T$

In $P_k(\mathbf{x}|y; \theta^S)$, leading to biased representation learning. To address this, we propose logit calibration by adaptively substituting the global prior in the teacher logits with the local prior, yielding the calibrated teacher logit and probability:

$$\mathbf{z}_y^{\text{cal}} = \mathbf{z}_y^T + \ln P_k^S(y) - \ln P_g^T(y) = \ln P_g(\mathbf{x}|y; \theta^T) + \ln P_k^S(y). \quad (11)$$

$$\mathbf{p}^{\text{cal}} = \text{Softmax}(\mathbf{z}^{\text{cal}}). \quad (12)$$

This operation removes the global prior bias while preserving the likelihood component from the teacher model. As a result, aligning student logits with calibrated teacher logits (i.e., $\mathbf{z}_y^S \approx \mathbf{z}_y^{\text{cal}}$) effectively reduces to matching the likelihood terms: $\ln P_k(\mathbf{x}|y; \theta^S) \approx \ln P_g(\mathbf{x}|y; \theta^T)$. This encourages the local student model to learn globally consistent and transferable features without being skewed by local data imbalance. Note that the server can securely aggregate client-side prior statistics through differential privacy [24] or homomorphic encryption [26] mechanisms, enabling estimation of the global prior distribution while preserving data confidentiality.

4.3 Adaptive Decoupled Knowledge Distillation

While calibration addresses prior mismatch, conventional KD still couples all class logits into a single KL divergence term, ignoring the heterogeneous importance of different class groups. Empirical evidence (e.g., Figure 2) reveals that under-represented classes suffer more severe forgetting than over-represented ones. To tackle this, we propose adaptive decoupled distillation, where the knowledge is

divided into three disjoint components to prioritize the preservation of critical knowledge components, particularly the fragile UCK. The overall local loss function of FedAKD is formulated as:

$$\begin{aligned} \mathcal{L}_{\text{FedAKD}} = & \mathcal{L}_{\text{CE}} + \underbrace{\text{KL}(\mathbf{p}_{\text{group}}^{\text{cal}} || \mathbf{p}_{\text{group}}^S)}_{\mathcal{L}_{\text{GRK}}} \\ & + \beta \cdot \underbrace{\text{KL}(\hat{\mathbf{p}}_{\text{over}}^{\text{cal}} || \hat{\mathbf{p}}_{\text{over}}^S)}_{\mathcal{L}_{\text{OCK}}} + \lambda \cdot \underbrace{\text{KL}(\hat{\mathbf{p}}_{\text{under}}^{\text{cal}} || \hat{\mathbf{p}}_{\text{under}}^S)}_{\mathcal{L}_{\text{UCK}}}. \end{aligned} \quad (13)$$

$\mathcal{L}_{\text{GRK}}$ preserves the teacher model's holistic inter-group confidence distribution, $\mathcal{L}_{\text{OCK}}$ maintains discriminative patterns among over-represented classes and preserves local dominance, while $\mathcal{L}_{\text{UCK}}$ mitigates local under-represented neglect. To counteract the suppression of UCK in traditional distillation, we introduce $\lambda \geq \beta$ as a prioritized weighting factor. This amplification ensures sustained emphasis on fine-grained knowledge of under-represented classes, effectively alleviating catastrophic forgetting caused by data scarcity.

In addition, the local cross-entropy loss $\mathcal{L}_{\text{CE}}$ inherently enables the student model to learn ground-truth label knowledge from the teacher model [9, 12], to avoid gradient conflicts between the $\mathcal{L}_{\text{CE}}$ and distillation loss, particularly in low-confidence predictions, we apply ground-truth masking. Specifically, we exclude the ground-truth class logit from both the calibrated teacher and the student outputs before computing the distillation loss.

5 Experiment

5.1 Experiment Setup

We conducted experiments on MNIST [7], CIFAR-10 [6], CIFAR-100 [6] and Tiny-Imagenet [22] datasets, and employ the Sharding (S) strategy [16] and Latent Dirichlet Allocation (LDA) strategy Dir(α) [15] to divide the dataset for each client. In terms of the global model, we utilized the SimpleCNN [8, 9, 12] on MNIST, CIFAR-10, and CIFAR-100 datasets, while employing MobileNet-V2 on Tiny-Imagenet. Our proposed approach FedAKD was compared with FedAvg (2017) [16], FedProx (2020) [11], FedCurv (2019) [20], MOON (2021) [10], FedNTD (2022) [9], FedLMD (2023) [12], FedSOL (2024) [8], and FedVLS (2025) [3]. Following prior studies [8, 9, 12] for fair comparison, we fixed the random seed for reproducible client selection and results.

Key parameters: 200 communication rounds (T), 100 clients (K) with 10% ($R=0.1$) sampled per round, 5 local epochs (E), batch size 50. Optimizer: SGD (initial learning rate $\eta_0=0.01$, momentum 0.9, weight decay 10^{-5}), learning rate decay $\eta_t = \eta_0 \times 0.99^t$.⁷

5.2 Overall Performance

Tables 1 and 2 demonstrate that our proposed FedAKD achieves state-of-the-art performance across varying degrees of data heterogeneity. Under pathological sharding non-IID settings, FedAKD improves accuracy over FedAvg by 9.44%, 4.86%, and 5.26% on CIFAR-10, CIFAR-100, and Tiny-Imagenet, respectively, and achieves about 1% higher performance than the best existing method. Under Dirichlet-based non-IID settings, the gains are 8.56%, 2.12%, and 3.22% on the

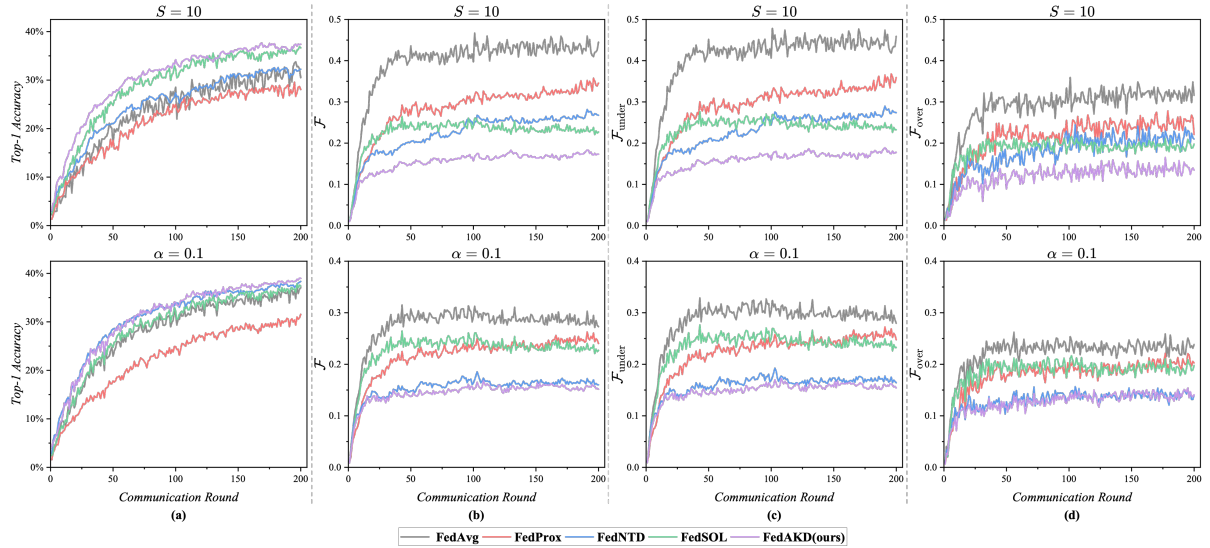
⁷Detailed experiment settings are provided in Appendix C.

Table 1: The Top-1 accuracy (%) of the global model under varied sharding strategies.

Method	MNIST	CIFAR-10				CIFAR-100		TinyImageNet	
	$S = 2$	$S = 2$	$S = 3$	$S = 5$	$S = 10$	$S = 10$	$S = 20$	$S = 20$	$S = 40$
FedAvg	95.35	49.18	57.41	68.65	71.89	32.16	34.96	34.52	40.31
FedProx	95.12	50.10	55.74	65.18	66.61	28.30	30.56	25.03	29.93
FedCurv	95.54	52.37	56.25	65.74	67.99	29.60	33.35	32.84	37.96
FedNova	93.25	44.92	54.31	64.93	68.59	29.18	31.83	32.17	38.95
MOON	95.87	49.18	55.90	68.62	72.14	32.11	33.74	35.40	40.62
FedNTD	96.70	63.04	67.62	72.85	74.69	36.67	38.02	38.62	43.13
FedLMD	97.02	63.54	68.12	73.21	75.37	<u>36.81</u>	<u>38.53</u>	39.21	<u>43.85</u>
FedSOL	96.98	63.27	65.73	<u>73.37</u>	74.18	36.19	37.01	35.37	40.71
FedVLS	<u>97.06</u>	<u>63.69</u>	<u>68.31</u>	72.64	74.59	35.64	36.56	<u>39.27</u>	42.78
FedAKD (Ours)	97.49	66.43	69.06	74.49	<u>74.91</u>	37.49	39.34	40.96	44.39

Table 2: The Top-1 accuracy (%) of the global model under varied LDA strategies.

Method	MNIST	CIFAR-10				CIFAR-100		TinyImageNet	
	$\alpha = 0.1$	$\alpha = 0.05$	$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.1$	$\alpha = 0.5$
FedAvg	96.15	39.19	54.78	66.16	70.78	36.39	37.31	39.06	44.08
FedProx	95.28	39.77	51.69	63.87	65.05	31.23	32.47	28.80	33.25
FedCurv	96.39	41.51	53.35	63.12	67.13	34.91	36.09	34.36	43.92
FedNova	86.75	10.00	35.91	61.36	67.28	33.25	34.68	38.78	42.68
MOON	96.03	39.07	54.65	64.59	70.98	36.37	37.82	39.22	40.16
FedNTD	96.61	47.29	63.81	68.88	72.21	<u>38.62</u>	39.11	41.51	45.19
FedLMD	96.96	49.31	63.76	70.22	<u>73.83</u>	38.34	39.06	41.03	<u>45.33</u>
FedSOL	<u>97.04</u>	50.12	65.36	70.47	72.11	37.84	<u>39.19</u>	39.65	41.28
FedVLS	96.95	52.47	<u>65.63</u>	<u>70.57</u>	73.42	36.83	37.66	<u>42.11</u>	45.13
FedAKD (Ours)	97.33	<u>52.32</u>	66.28	72.01	74.53	38.66	39.27	43.53	46.05

**Figure 4: The Top-1 accuracy, $\mathcal{F}$, $\mathcal{F}_{\text{under}}$, and $\mathcal{F}_{\text{over}}$ of different approaches on the CIFAR-100 dataset under different rounds.**

three datasets, also outperforming the state-of-the-art by 0.6%–1.1%. Meanwhile, we compared the training processes of different methods on the CIFAR-100 dataset ($S = 10$, $\alpha = 0.1$). As shown in Figure 4 (a), our method demonstrates both faster convergence and greater stability compared to baseline methods.

In addition, we recorded knowledge forgetting metrics $\mathcal{F}$, $\mathcal{F}_{\text{under}}$, and $\mathcal{F}_{\text{over}}$ for different methods during the training process on the CIFAR-100 dataset ($S = 10$, $\alpha = 0.1$). As illustrated in Figures 4 (b)–(d), our method maintains a consistently lower, more stable degree of knowledge forgetting throughout training. Notably, it exhibits a greater reduction in $\mathcal{F}_{\text{under}}$ than $\mathcal{F}_{\text{over}}$, suggesting that reducing the forgetting of the under-represented class knowledge has a more pronounced impact on improving FL performance.

5.3 Ablation Study

We conducted ablation studies on the CIFAR-10 dataset to verify the contribution of the Adaptive Knowledge Calibration (AKC) and the Adaptive Decoupled Knowledge Distillation (ADKD). As shown in Table 3, the results confirm that both AKC and ADKD modules significantly enhance performance, validating the efficacy of the proposed FedAKD.

Table 3: The Top-1 accuracy of the global model on the CIFAR-10 dataset under varied module combinations.

Partitioning Strategy	AKC	ADKD	Acc.(%)
$S = 2$	✗	✗	53.18
	✓	✗	65.66 (+12.49)
	✓	✓	66.43 (+13.26)
$\alpha = 0.1$	✗	✗	56.41
	✓	✗	65.72 (+9.31)
	✓	✓	66.28 (+9.87)

5.4 Hyperparameter Analysis

We conducted a parameter sensitivity analysis on the CIFAR-10 dataset to assess the impact of hyperparameters β and λ on model performance in FedAKD. Specifically, we independently analyzed each parameter: fixing $\lambda = 1$ to explore the optimization boundaries of β , and fixing $\beta = 1$ to evaluate the influence of λ . The experimental result is shown in Figure 5, demonstrating that the global model performance improves when the hyperparameter λ is greater than or equal to β , whereas the model performance degrades when λ is less than β . This finding aligns with our previous analysis. Therefore, we selected $\beta = 1$ and $\lambda = 2$ for the experiments on the pathological sharding distribution, and $\beta = 0.5$ and $\lambda = 1$ for the experiments on the LDA distribution.

5.5 The Impact of Local Training Epochs

We explored how varying numbers of local training rounds ($E \in \{1, 3, 5, 10\}$) affect the final global model performance on the CIFAR-10 dataset. As demonstrated in Figure 6, FedAKD achieves superior performance across varying local training rounds (E), sustaining strong accuracy even under extreme parameterizations ($E = 1$) in both sharding ($S = 2$) and LDA ($\alpha = 0.1$) non-IID scenarios.

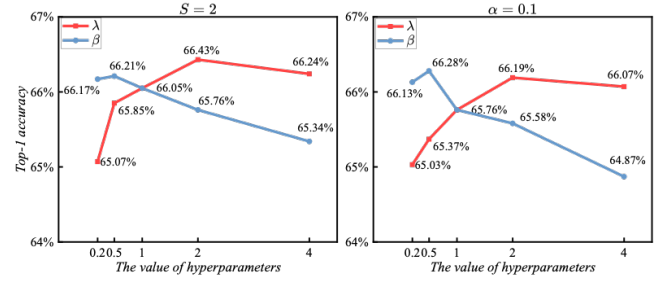


Figure 5: Hyperparameter analysis.

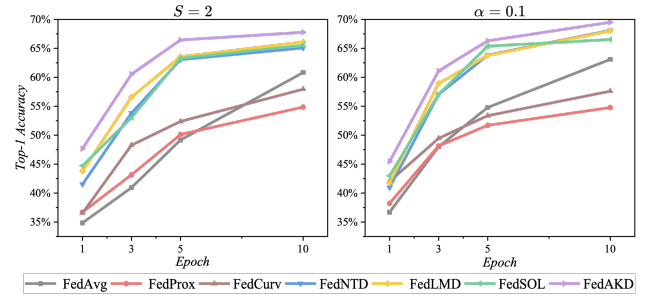


Figure 6: The impact of local training epochs on the global model performance.

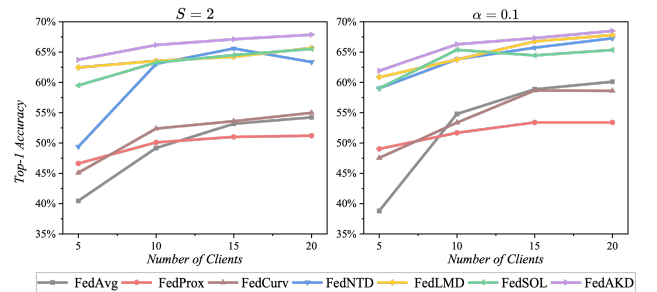


Figure 7: The impact of client participation rate on the global model performance.

Scaling E from 1 to 10 yields the most substantial performance improvements, empirically validating FedAKD’s robustness to heterogeneous training configurations and reflecting that FedAKD can effectively alleviate the phenomenon of global knowledge forgetting during the local training.

5.6 The Impact of Client Participation Rate

We investigated the impact of client participation rates (R) on the global model performance on the CIFAR-10 dataset, where $R \in \{0.05, 0.1, 0.15, 0.2\}$ (equivalent to the number of participating clients $m \in \{5, 10, 15, 20\}$). As shown in Figure 7, the proposed FedAKD demonstrates relatively superior performance across heterogeneous participation scenarios, highlighting its robustness under different client participation rates. This may be due to its ability to preserve the global knowledge effectively.

Table 4: The Top-1 accuracy (%) of the global model under different network architectures.

Method	$S = 2$				$\alpha = 0.1$			
	Simple CNN	ResNet 10	VGG 11	Small VIT	Simple CNN	ResNet 10	VGG 11	Small VIT
FedAvg	49.18	43.32	33.81	33.53	54.78	52.73	39.91	33.65
FedProx	50.10	49.20	37.83	37.83	51.69	59.41	40.30	39.86
FedCurv	52.37	45.69	35.86	38.79	53.35	52.22	43.09	38.98
FedNTD	63.04	57.73	49.27	43.49	<u>63.81</u>	60.94	57.55	43.53
FedLMD	<u>63.54</u>	<u>58.18</u>	<u>54.40</u>	<u>44.78</u>	63.76	<u>61.91</u>	<u>59.27</u>	<u>43.97</u>
FedAKD (Ours)	66.43	61.84	55.93	48.62	66.28	62.95	62.42	46.29

5.7 Comparative Analysis of Different Global Model Architectures.

We verified the applicability of our FedAKD with different global model architectures and illustrated the performance with several typical architectures on the CIFAR-10 dataset. As indicated in Table 4, the proposed FedAKD outperforms other methods across all network architectures. Specifically, in the sharding scenario, FedAKD achieves an average cross-network Top-1 accuracy of 58.21%, surpassing the second-best method by 2.98%. In the LDA scenario, FedAKD attains an average cross-network accuracy of 59.49%, a 2.26% improvement over FedLMD. These results confirm FedAKD’s adaptability to diverse global model architectures.

5.8 Efficiency Analysis

The adaptive knowledge calibration module in FedAKD introduces two sources of additional computation: (i) a one-time calculation of the local class prior $P_k(y)$, and (ii) logit calibration according to $z_y^{\text{cal}} = z_y^T + \log P_k(y) - \log P_g(y)$. Both steps are vector-level operations with a complexity of $O(C)$, where C denotes the number of classes. Moreover, during the knowledge distillation of GRK, OCK, and UCK, the computation of KL divergence has the same complexity as standard knowledge distillation, i.e., $O(C)$. Therefore, compared with conventional KL-based distillation, the decoupled distillation mechanism introduces no additional computational burden, and its overhead is negligible relative to backbone model training. It is important to emphasize that FedAKD incurs no extra computational cost during inference. In terms of communication, each client only uploads an additional class-prior vector of size C , which is negligible compared to model updates.

To quantify the additional overhead during local training, we conducted a comparative analysis on the CIFAR-100 dataset under a non-IID data partition ($\alpha = 0.1$) with a client participation rate of $R = 0.1$. As detailed in Table 5, FedAKD achieves superior accuracy with only a negligible increase in resource consumption. Specifically, it boosts the Top-1 accuracy to 38.66% while incurring only a marginal 2.1% increase in training time compared to FedAvg. Notably, the required GPU memory remains identical to that of FedAvg and is considerably lower than methods like FedCurv. This empirical evidence confirms that FedAKD provides a substantial performance gain without significant computational or memory overhead, making it a practical and efficient solution for resource-constrained Web, mobile, and WoT intelligence deployment.

Table 5: Performance and resource cost comparison of different federated learning methods.

Method	Acc. (%)	Time (s)	Memory (MiB)
FedAvg	36.39	136.8	2546
FedProx	31.23	140.3	2546
FedCurv	34.91	147.6	3280
FedNova	33.25	164.0	2586
MOON	36.37	154.6	2566
FedNTD	38.62	137.2	2546
FedLMD	38.34	138.9	2546
FedSOL	37.84	153.8	2546
FedVLS	36.83	152.5	2546
FedAKD (Ours)	38.66	139.7	2546

6 Conclusion

In this work, we demonstrated that existing client-side federated KD methods overlook the differences between local and global prior distributions, causing local models to learn distorted global knowledge with implicit local prior biases. Moreover, we found that local over- and under-represented classes exhibit varying degrees of knowledge forgetting, while traditional KD methods cannot selectively and flexibly retain global knowledge. To overcome these challenges, we proposed FedAKD, an innovative methodology integrating adaptive logit calibration and decoupled distillation simultaneously. The logit calibration prior eliminates the inherent bias between global and local knowledge, while decoupled knowledge distillation enables selective and purposeful knowledge retention to accommodate heterogeneous data environments. Extensive experiments confirm that the proposed FedAKD significantly outperforms state-of-the-art methods under various non-IID conditions, which provides a practical and scalable solution for intelligent model collaboration across Web, mobile, and WoT systems.

Limitations: The FedAKD framework currently relies on client-reported priors for global estimation, requiring stronger security guarantees against unreliable or malicious participants. Additionally, the distillation weights are static hyperparameters, limiting adaptive optimization under complex or evolving non-IID data distributions.

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A Proof of Theorem 1

Construct the following hypothesis testing problem: (1) The null hypothesis H_0 : The local proportion $p_k^{(i)}$ is not significantly higher than the global proportion p , i.e., $p_k^{(i)} \leq p$. (2) The alternative hypothesis H_1 : The local proportion $p_k^{(i)}$ is significantly higher than the global proportion p , i.e., $p_k^{(i)} > p$.

According to the Central Limit Theorem, which states that under certain conditions, the distribution of the sum of a large number of independent random variables will approach a normal distribution, when the sample size n_k is sufficiently large, typically requiring $n_k \cdot p \geq 10$ and $n_k \cdot (1-p) \geq 10$, the distribution of the sample mean (or proportion) will approach a normal distribution. Specifically, for a binomial distribution $B(n_k, p)$, the sample proportion $p_k^{(i)}$ will approximately follow a normal distribution: $p_k^{(i)} \sim N\left(p, \frac{p(1-p)}{n_k}\right)$. Therefore, the test statistic Z is given by:

$$Z = \frac{p_k^{(i)} - p}{\sqrt{\frac{p(1-p)}{n_k}}} \quad (14)$$

For a significance level δ , the critical value z_δ for a one-sided test satisfies $P(Z \geq z_\delta) = \delta$ (for example, when $\delta = 0.05$, $z_\delta \approx 1.645$). In other words, if the test statistic Z satisfies $Z \geq z_\delta$, we reject the null hypothesis H_0 and conclude that the local proportion $p_k^{(i)}$ is significantly higher than the global proportion p .

Deriving the lower bound of ν . Based on the above, substituting the relative difference ratio $\nu = \frac{p_k^{(i)} - p}{p}$ into the expression for the test statistic in Eq. (14), we obtain:

$$Z = \frac{\nu \cdot p}{\sqrt{\frac{p(1-p)}{n_k}}} \quad (15)$$

Given the rejection region condition $Z \geq z_\delta$, we have:

$$\frac{\nu \cdot p}{\sqrt{\frac{p(1-p)}{n_k}}} \geq z_\delta \quad (16)$$

Rearranging the inequality yields:

$$\nu \geq z_\delta \cdot \sqrt{\frac{1-p}{n_k p}} \quad (17)$$

which completes the proof.

B Proof of Proposition 1

Let the output of the teacher model be $\mathbf{p}^\mathcal{T}$ and the output of the student model be $\mathbf{p}^\mathcal{S}$, where $\mathcal{T}$ and $\mathcal{S}$ denote the teacher and student models, respectively. The distillation loss is based on the Kullback-Leibler (KL) divergence. For simplicity, the distillation temperature τ is omitted here. The KL divergence between $\mathbf{p}^\mathcal{T}$ and $\mathbf{p}^\mathcal{S}$ is:

$$\text{KL}(\mathbf{p}^\mathcal{T} \parallel \mathbf{p}^\mathcal{S}) = \sum_{i=1}^C p_i^\mathcal{T} \log \frac{p_i^\mathcal{T}}{p_i^\mathcal{S}} = \sum_{i=1}^l p_i^\mathcal{T} \log \frac{p_i^\mathcal{T}}{p_i^\mathcal{S}} + \sum_{i=l+1}^C p_i^\mathcal{T} \log \frac{p_i^\mathcal{T}}{p_i^\mathcal{S}}$$

where $i \leq l$ represents the over-represented classes and $i > l$ represents the under-represented classes.

Using Definition 3.3, we substitute $p_i^\mathcal{T}$ and $p_i^\mathcal{S}$ with their hierarchical forms, yielding:

$$\begin{aligned} & \text{KL}(\mathbf{p}^\mathcal{T} \parallel \mathbf{p}^\mathcal{S}) \\ &= \sum_{i=1}^l \left(p_{\text{over}}^\mathcal{T} \hat{p}_i^\mathcal{T} \log \frac{p_{\text{over}}^\mathcal{T} \hat{p}_i^\mathcal{T}}{p_{\text{over}}^\mathcal{S} \hat{p}_i^\mathcal{S}} \right) + \sum_{i=l+1}^C \left(p_{\text{under}}^\mathcal{T} \check{p}_i^\mathcal{T} \log \frac{p_{\text{under}}^\mathcal{T} \check{p}_i^\mathcal{T}}{p_{\text{under}}^\mathcal{S} \check{p}_i^\mathcal{S}} \right) \\ &= \sum_{i=1}^l p_{\text{over}}^\mathcal{T} \hat{p}_i^\mathcal{T} \left(\log \frac{p_{\text{over}}^\mathcal{T}}{p_{\text{over}}^\mathcal{S}} + \log \frac{\hat{p}_i^\mathcal{T}}{\hat{p}_i^\mathcal{S}} \right) \\ &\quad + \sum_{i=l+1}^C p_{\text{under}}^\mathcal{T} \check{p}_i^\mathcal{T} \left(\log \frac{p_{\text{under}}^\mathcal{T}}{p_{\text{under}}^\mathcal{S}} + \log \frac{\check{p}_i^\mathcal{T}}{\check{p}_i^\mathcal{S}} \right) \\ &= \underbrace{\sum_{i=1}^l p_{\text{over}}^\mathcal{T} \hat{p}_i^\mathcal{T} \log \frac{p_{\text{over}}^\mathcal{T}}{p_{\text{over}}^\mathcal{S}} + \sum_{i=l+1}^C p_{\text{under}}^\mathcal{T} \check{p}_i^\mathcal{T} \log \frac{p_{\text{under}}^\mathcal{T}}{p_{\text{under}}^\mathcal{S}}}_{\text{GRK Distillation}} \\ &\quad + \underbrace{p_{\text{over}}^\mathcal{T} \sum_{i=1}^l \hat{p}_i^\mathcal{T} \log \frac{\hat{p}_i^\mathcal{T}}{\hat{p}_i^\mathcal{S}}}_{\text{OCK Distillation}} + \underbrace{p_{\text{under}}^\mathcal{T} \sum_{i=l+1}^C \check{p}_i^\mathcal{T} \log \frac{\check{p}_i^\mathcal{T}}{\check{p}_i^\mathcal{S}}}_{\text{UCK Distillation}} \\ &= p_{\text{over}}^\mathcal{T} \log \frac{p_{\text{over}}^\mathcal{T}}{p_{\text{over}}^\mathcal{S}} \sum_{i=1}^l \hat{p}_i^\mathcal{T} + p_{\text{under}}^\mathcal{T} \log \frac{p_{\text{under}}^\mathcal{T}}{p_{\text{under}}^\mathcal{S}} \sum_{i=l+1}^C \check{p}_i^\mathcal{T} \\ &\quad + p_{\text{over}}^\mathcal{T} \cdot \text{KL}(\hat{\mathbf{p}}_{\text{over}}^\mathcal{T} \parallel \hat{\mathbf{p}}_{\text{over}}^\mathcal{S}) + p_{\text{under}}^\mathcal{T} \cdot \text{KL}(\check{\mathbf{p}}_{\text{under}}^\mathcal{T} \parallel \check{\mathbf{p}}_{\text{under}}^\mathcal{S}) \\ &= p_{\text{over}}^\mathcal{T} \log \frac{p_{\text{over}}^\mathcal{T}}{p_{\text{over}}^\mathcal{S}} + p_{\text{under}}^\mathcal{T} \log \frac{p_{\text{under}}^\mathcal{T}}{p_{\text{under}}^\mathcal{S}} \\ &\quad + p_{\text{over}}^\mathcal{T} \cdot \text{KL}(\hat{\mathbf{p}}_{\text{over}}^\mathcal{T} \parallel \hat{\mathbf{p}}_{\text{over}}^\mathcal{S}) + p_{\text{under}}^\mathcal{T} \cdot \text{KL}(\check{\mathbf{p}}_{\text{under}}^\mathcal{T} \parallel \check{\mathbf{p}}_{\text{under}}^\mathcal{S}) \\ &= \underbrace{\text{KL}(\mathbf{p}_{\text{inter}}^\mathcal{T} \parallel \mathbf{p}_{\text{inter}}^\mathcal{S})}_{\text{GRK Distillation}} + \underbrace{p_{\text{over}}^\mathcal{T} \cdot \text{KL}(\hat{\mathbf{p}}_{\text{over}}^\mathcal{T} \parallel \hat{\mathbf{p}}_{\text{over}}^\mathcal{S})}_{\text{OCK Distillation}} \\ &\quad + \underbrace{p_{\text{under}}^\mathcal{T} \cdot \text{KL}(\check{\mathbf{p}}_{\text{under}}^\mathcal{T} \parallel \check{\mathbf{p}}_{\text{under}}^\mathcal{S})}_{\text{UCK Distillation}} \end{aligned} \quad (18)$$

which completes the proof.

C Experimental Details

C.1 Datasets

To ensure a fair comparison, we selected four widely-used datasets in the federated learning domain for our experiments: MNIST, CIFAR-10, CIFAR-100, and Tiny-ImageNet. The MNIST dataset, a classic benchmark for image recognition, consists of 60,000 grayscale training images and 10,000 test images of handwritten digits, each with a resolution of 28×28 pixels. The CIFAR-10 dataset contains 60,000 32×32 color images, divided into 10 classes, with 50,000 images for training and 10,000 for testing. The CIFAR-100 dataset, which shares a similar structure with CIFAR-10, includes 100 classes, each comprising 600 images (500 for training and 100 for testing). The Tiny-ImageNet dataset, a subset of ImageNet, encompasses 200 classes, with 500 training images, 50 validation images, and 50 test images per class, totaling approximately 100,000 images. For data augmentation, we applied techniques such as random cropping, random horizontal flipping, and normalization to the training data, consistent with prior studies.

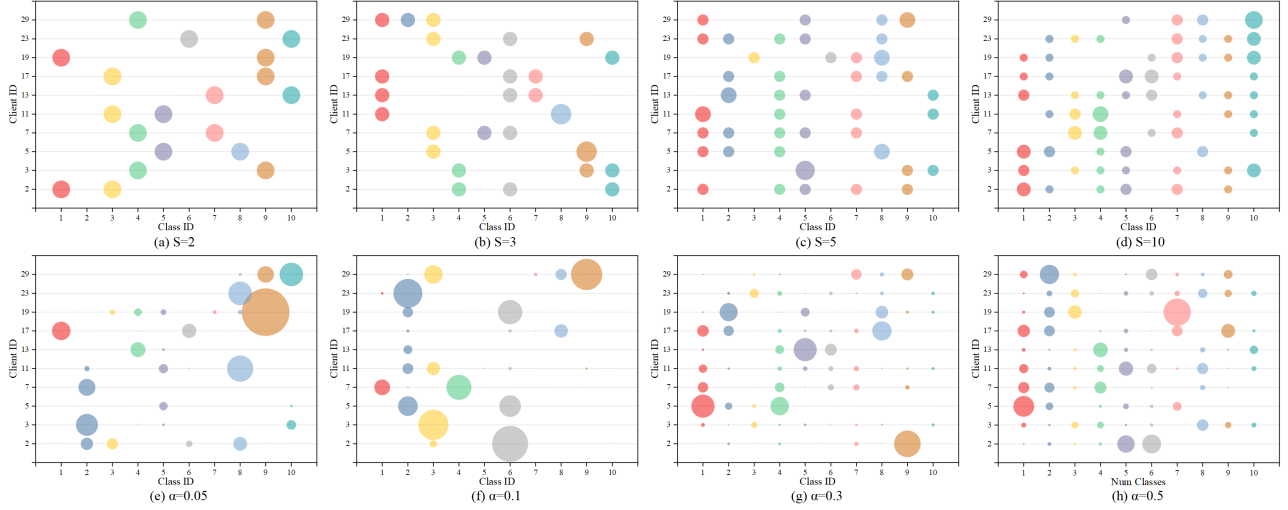


Figure 8: Two distinct partitioning strategies were utilized to segment the CIFAR-10 dataset, with visualizations depicting the distribution of different clients and their corresponding datasets. Different colors in the visualizations signify different categories within the dataset. For illustrative purposes, we randomly selected 10 clients. Here, S denotes the dataset partitioned using the pathological sharding strategy, and α represents the dataset partitioned using the LDA strategy.

C.2 Data Partitioning Strategies

To simulate the non-independent and identically distributed (Non-IID) scenario of client data in federated learning, we employed two data partitioning strategies: the pathological distribution strategy and the Dirichlet distribution strategy. The pathological distribution strategy considers only statistical heterogeneity, as all clients have datasets of the same size, and there is no overlap between the data samples of different clients. In this pathological distribution, for a given dataset, the smaller the parameter S , the fewer categories each client covers, the more skewed the local data label distribution becomes, and the stronger the global data heterogeneity is. Under the Dirichlet distribution strategy, each client's dataset varies in size and category distribution, potentially containing major categories, minor categories, or even missing some categories, thus more closely approximating real-world federated learning data scenarios. Specifically, for the MNIST dataset, we set $S = 2$ and $\alpha = 0.1$. For the CIFAR-10 dataset, we set $S \in \{2, 3, 5, 10\}$ and $\alpha \in \{0.05, 0.1, 0.3, 0.5\}$. For the CIFAR-100 dataset, we set $S \in \{10, 20\}$ and $\alpha \in \{0.1, 0.5\}$. For the Tiny-ImageNet dataset, we set $S \in \{10, 20\}$ and $\alpha \in \{0.1, 0.5\}$. Figure 8 illustrates the skewed data distribution resulting from these different data partitioning strategies on the CIFAR-10 dataset.

C.3 Experimental Environment

The experiments were conducted using the PyTorch framework, version 1.11.0. All experiments were completed on a single NVIDIA

GeForce 4090 GPU with 24GB of memory. The overall code structure was based on the FedNTD library, which was modified and implemented accordingly.

C.4 Experimental Parameter Settings for Comparison Methods

To ensure fairness, the hyperparameters for different methods were adjusted according to the configurations specified in the original papers. Specific values are detailed in Tables 6 and 7.

D On the Choice and Sensitivity of the Threshold ν

To empirically validate the robustness of our approach to the choice of the ν threshold, we conducted a sensitivity analysis. We varied the confidence level from 90% to 99% and observed the impact on the final Top-1 accuracy of our model. The experiment was performed on the CIFAR-100 dataset under a Dirichlet distribution with $\alpha = 0.1$ and a 10% client participation rate.

The results are presented in Table 8, which demonstrates remarkable stability. Across a wide range of confidence levels, the Top-1 accuracy remains consistently high, with a mean of 38.47% and a low standard deviation of only 0.27%. The total fluctuation is minimal, with a maximum accuracy of 38.72% and a minimum of 37.89%. This low sensitivity confirms that our method is robust to the exact choice of ν and that the theoretically-derived lower bound (e.g., at a 95% confidence level) is sufficient for effective performance.

Table 6: Hyperparameter settings for the pathological sharding Non-IID distribution (S).

Method	MNIST	CIFAR-10	CIFAR-100	Tiny-ImageNet
FedCurv	$s = 500, \lambda = 2$	$s = 250, \lambda = 2 (S = 2)$ $s = 500, \lambda = 1 (S = 3, 5, 10)$	$s = 500, \lambda = 1 (S = 10)$ $s = 500, \lambda = 2 (S = 20)$	$s = 500, \lambda = 1 (S = 20)$ $s = 500, \lambda = 2 (S = 40)$
FedProx	$\mu = 0.5$			
FedNOVA	None			
MOON	$\mu = 0.1, \tau = 1$	$\mu = 0.1, \tau = 1 (S = 2, 3, 5)$ $\mu = 0.1, \tau = 0.5 (S = 10)$	$\mu = 0.1, \tau = 1$	
FedNTD	$\beta = 1, \tau = 1$	$\beta = 3, \tau = 1 (S = 2, 5)$ $\beta = 1, \tau = 1 (S = 3, 10)$	$\beta = 3, \tau = 3 (S = 10)$ $\beta = 1, \tau = 3 (S = 20)$	$\beta = 3, \tau = 3 (S = 20)$ $\beta = 1, \tau = 3 (S = 40)$
FedLMD	$\beta = 1, \tau = 1$		$\beta = 2, \tau = 3 (S = 10)$ $\beta = 1, \tau = 3 (S = 20)$	$\beta = 2, \tau = 3 (S = 20)$ $\beta = 1, \tau = 3 (S = 40)$
FedSOL	$\rho = 2, \tau = 1$		$\rho = 2, \tau = 3$	
FedVLS	$\lambda = 1, \tau = 1$		$\lambda = 1, \tau = 3$	
FedAKD	$\beta = 1, \lambda = 2, \tau = 1$		$\beta = 1, \lambda = 2, \tau = 3$	

Table 7: Hyperparameter settings for the Dirichlet Non-IID distribution ($\text{Dir}(\alpha)$).

Method	MNIST	CIFAR-10	CIFAR-100	Tiny-ImageNet
FedCurv	$s = 500, \lambda = 2$	$s = 250, \lambda = 2 (\alpha = 0.05)$ $s = 500, \lambda = 1 (\alpha = 0.1, 0.3, 0.5)$	$s = 500, \lambda = 1 (\alpha = 0.1)$ $s = 500, \lambda = 2 (\alpha = 0.5)$	$s = 500, \lambda = 1 (\alpha = 0.1)$ $s = 500, \lambda = 2 (\alpha = 0.5)$
FedProx	$\mu = 0.5$			
FedNOVA	None			
MOON	$\mu = 0.1, \tau = 1$	$\mu = 0.1, \tau = 1 (\alpha = 0.05, 0.1, 0.3)$ $\mu = 0.1, \tau = 0.5 (\alpha = 0.5)$	$\mu = 0.1, \tau = 1$	
FedNTD	$\beta = 1, \tau = 1$	$\beta = 3, \tau = 1 (\alpha = 0.05)$ $\beta = 1, \tau = 1 (\alpha = 0.1)$ $\beta = 2, \tau = 1 (\alpha = 0.3, 0.5)$	$\beta = 3, \tau = 1$	
FedLMD	$\beta = 1, \tau = 1$	$\beta = 2, \tau = 3 (\alpha = 0.05)$ $\beta = 1, \tau = 1 (\alpha = 0.1, 0.3, 0.5)$	$\beta = 1, \tau = 1$	
FedSOL	$\rho = 2, \tau = 1$	$\rho = 2, \tau = 1$	$\rho = 2, \tau = 3$	
FedVLS	$\lambda = 1, \tau = 1$	$\lambda = 1, \tau = 1$	$\lambda = 1, \tau = 3$	
FedAKD	$\beta = 0.5, \lambda = 1, \tau = 1$	$\beta = 0.5, \lambda = 1, \tau = 1$	$\beta = 0.5, \lambda = 1, \tau = 3$	

Table 8: Sensitivity analysis of Top-1 accuracy (%) with respect to the confidence level used for threshold ν . Experiments were run on CIFAR-100 with a Dirichlet distribution ($\alpha = 0.1$).

Confidence Level	Significance (δ)	Z-value (z_δ)	Threshold (ν)	Top-1 Accuracy (%)
90%	0.10	1.282	0.57	37.89
91%	0.09	1.341	0.60	38.11
92%	0.08	1.405	0.63	38.45
93%	0.07	1.476	0.66	38.45
94%	0.06	1.555	0.69	38.49
95%	0.05	1.645	0.73	38.66
96%	0.04	1.751	0.78	38.59
97%	0.03	1.881	0.84	38.72
98%	0.02	2.054	0.91	38.67
99%	0.01	2.326	1.04	38.63